

# Association between heterozygous *GBA1* L444P carrier status and risk of Parkinson's disease: A systematic review and meta-analysis

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## ABSTRACT

**Introduction and objective:** Mutations in the *GBA1* gene, particularly the severe L444P variant, are among the strongest known genetic risk factors for Parkinson's disease (PD). Prior meta-analyses published before 2018 included fewer studies and often pooled multiple mutations, prompting this updated, mutation-specific systematic review and meta-analysis.

**Methods:** Standard systematic review and meta-analytic methods were used, including predefined eligibility criteria, literature search, and pooled analysis of odds ratios (ORs) for PD in *GBA1* L444P carriers versus non-carriers.

**Results:** We included 51 studies comprising 33,752 PD patients and 34,101 controls across multiple continents. In random-effects meta-analysis, heterozygous *GBA1* L444P carriers had markedly higher odds of PD than non-carriers (pooled OR 9.19, 95% CI 6.94–12.16), with similar estimates across sensitivity and leave-one-out analyses. Cumulative meta-analysis showed early variability but stable pooled ORs around 7–9 from 2008 to 2010 onward. Ancestry-specific pooled ORs were consistently elevated across populations, with overlapping confidence intervals and no significant subgroup differences. Funnel and Galbraith plots and Egger's test showed no evidence of substantial small-study effects or publication bias.

**Conclusions:** This updated synthesis demonstrates that heterozygous *GBA1* L444P carriage confers a consistently large increase in PD risk across populations, with robust pooled estimates across multiple sensitivity analyses. Although overall statistical heterogeneity was negligible, wide confidence intervals in several ancestry subgroups highlight imprecision due to sparse data and internal heterogeneity, underscoring the need for broader and more equitable genetic testing, improved risk estimation in underrepresented populations, and targeted research on severe *GBA1* variants to inform genetic counseling and PD risk discussions.

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## 1. Introduction

### 1.1. Rationale

*GBA1* encodes  $\beta$ -glucocerebrosidase (GCase), a lysosomal enzyme that degrades glucosylceramide [1]. GCase deficiency leads to accumulation of glucosylceramide and glucosylsphingosine in macrophages and neurons, leading to the multisystemic manifestations of Gaucher disease [2–5]. The link between *GBA1* mutations and neurodegenerative

disease was first noted over 30 years ago, when case reports described parkinsonian symptoms in individuals with type 1 Gaucher disease and, subsequently, in obligate and confirmed *GBA1* mutation carriers [6–8]. Subsequent large-scale sequencing and biobank analyses have shown that *GBA1* mutation carriers comprise roughly 5–20% of Parkinson's disease (PD) patients worldwide, making *GBA1* the most common genetic risk factor for PD and related Lewy body disorders [9]. *GBA1* variants are commonly classified as “mild,” “severe,” or “risk” variants based on their association with GD phenotype in homozygous or

**Abbreviations:** PD, Parkinson's disease; *GBA1*, glucocerebrosidase 1 gene; GCase,  $\beta$ -glucocerebrosidase; OR, odds ratio; CI, confidence interval; NOS, Newcastle–Ottawa Scale; REML, restricted maximum likelihood; MH, Mantel–Haenszel; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses;  $I^2$ ,  $I^2$  statistic;  $\tau^2$ , between-study variance; SE, standard error; Lyso-Gb1, glucosylsphingosine.

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compound heterozygous individuals [10], with this framework subsequently applied to heterozygous carriers in PD [11,12]. Variants associated with neuronopathic GD (Types 2 and 3), such as L444P (p. Leu483Pro; also referred to as L483P under Human Genome Variation Society nomenclature), are generally categorized as severe and are linked to markedly reduced  $\beta$ -glucocerebrosidase activity [13,14]. In contrast, variants such as N370S (p. Asn409Ser; also referred to as N409S), which are associated with non-neuronopathic GD, are considered mild [12]. Risk variants, including E326K (p. Glu365Lys; E365K) and T369M (p. Thr408Met; T408M), increase PD susceptibility but are not causative for GD [15,16]. Severe *GBA1* variants have consistently been associated with higher PD risk and earlier age at onset compared to mild variants, supporting a biologically graded effect of mutation severity [11,12,17]. Recent stratified analyses have demonstrated a clear genotype–phenotype gradient: severe mutations such as L444P (more commonly referred to as L483P by the Human Genome Variation Society nomenclature) or D448H are associated with a 9- to 10-fold higher PD risk, whereas mild variants like N409S are associated with a 3- to 4-fold higher risk [18,19].

Previous meta-analyses, including those by Mao et al. and Zhang et al., provided important evidence that *GBA1* variants are strongly associated with PD risk [20,21]. However, these analyses were based on fewer studies, covered literature up to 2013 and 2018 respectively, and often combined multiple *GBA1* mutations rather than evaluating L444P specifically. Since 2018, many additional genetic association and sequencing studies have been published, including larger case-controls and more diverse ancestry groups.

In this context, the purpose of this systematic review and meta-analysis is to incorporate recent data and provide a comprehensive and methodologically consistent estimate of the association between heterozygous *GBA1* L444P carrier status and PD risk. We focused on the L444P variant because it is a severe *GBA1* mutation associated with high PD risk and, unlike classic founder variants, it is present across multiple ancestries rather than being restricted to a single ethnic group. Our analytic framework follows established evidence synthesis principles: (1) a clearly defined population and exposure, (2) standardized extraction of effect measures, (3) evaluation of study quality and risk of bias, (4) random-effects meta-analysis to account for expected between-study variability, and (5) extensive planned and post hoc sensitivity analyses to evaluate heterogeneity and the stability of findings.

## 1.2. Research question

In observational genetic association studies of individuals with and without PD, does heterozygous *GBA1* L444P carrier status, compared with non-carrier status, increase the risk of PD?

## 1.3. Hypotheses

Based on prior studies of *GBA1* and PD, we expected heterozygous *GBA1* L444P carriers to have a substantially higher risk of PD than non-carriers. We also expected some variation in effect sizes across studies, driven by differences in ancestry, study size, genotyping methods, and diagnostic criteria, but anticipated that the overall direction of association would remain consistently positive.

## 2. Materials and methods

We followed contemporary guidance for systematic reviews, including dual screening, predefined eligibility criteria, structured data extraction, and risk-of-bias assessment [22].

### 2.1. Eligibility criteria

Inclusion criteria were: (i) an original human research article; (ii) a case–control or cross-sectional genetic association study evaluating the

association between heterozygous *GBA1* L444P carrier status and PD risk; (iii) PD diagnosed according to accepted clinical criteria (e.g., United Kingdom Brain Bank or Movement Disorder Society criteria); (iv) sufficient data, such as genotype counts, carrier frequencies, or reported effect estimates, to calculate odds ratios (ORs) and 95% confidence intervals (CIs); (v) publication in English. Studies were excluded if they: (i) contained overlapping or duplicate data, in which case the most complete or recent dataset was retained; (ii) were without a non-PD control group; (iii) were abstracts, comments, reviews, editorials, case reports, or case series; (iv) contained incomplete data such that ORs and 95% CIs could not be derived; (v) did not provide variant-specific results for L444P or only reported homozygous or compound heterozygous *GBA1* variants without extractable heterozygous L444P data.

### 2.2. Literature search

Published studies were identified by systematically searching PubMed and Embase.com from inception through October 16, 2025 (limited to articles published in English), using specified terms as keywords and subject headings (MeSH/Emtree) where applicable and combining them with Boolean operators. The key words used for the search included ‘*GBA1*’, ‘*GBA*’, and ‘glucocerebrosidase’ in combination with ‘Parkinson’s disease’, or parkinson\*, and with ‘case control study’ or ‘genetic association study’. In addition, reference lists of relevant articles were manually screened, forward citation searches were performed, and conference proceedings were searched. All searched papers were imported into Covidence for further eligibility evaluation. Two reviewers independently screened titles and abstracts against eligibility criteria. Full texts of potentially eligible studies were retrieved and reviewed in detail by the same two reviewers. Disagreements regarding study inclusion were resolved by a third reviewer.

### 2.3. Data extraction

For each eligible study, we collected information on first author, year of publication, country or region, ancestry or ethnicity of participants, study design, genotyping method, diagnostic criteria for PD, number of cases and controls, and the number of heterozygous *GBA1* L444P carriers in each group. When studies included subjects from different ancestry groups, data were extracted separately for each group whenever possible. If more than one paper used the same or overlapping case series, the publication with the most complete or recent dataset was included. Data were extracted manually by a primary human reviewer using a standardized form. As an additional quality-assurance step, a large language model-based assistant (ChatGPT, GPT-4 series, OpenAI) performed a parallel extraction. The two extractions were compared, and any discrepancies were checked against the original articles and resolved by a second human reviewer. The final dataset used for all analyses were based solely on human adjudication. Further details on AI-assisted procedures are provided in the Supplementary Methods.

### 2.4. Risk of bias assessment

Risk of bias for each included study was assessed using the Newcastle–Ottawa Scale (NOS), a widely used tool for evaluating the quality of case–control studies based on selection, comparability, and outcome/exposure. [23] NOS ratings were first assigned by a primary human reviewer and independently in parallel by a large language model-based assistant (ChatGPT, GPT-4 series, OpenAI) using the same predefined criteria. The two assessments were manually compared, and any discrepancies were checked against the original articles and resolved by a second human reviewer. Studies with higher NOS scores were considered to have lower risk of bias, whereas studies with lower scores were interpreted as having a higher potential for bias.

## 2.5. Effect measures

The association between heterozygous *GBA1* L444P carrier status and PD risk was quantified using ORs with 95% CIs. For each study, we constructed a  $2 \times 2$  table of L444P carrier status (carrier vs non-carrier) in cases and controls to maintain comparability across studies and avoid mixing adjusted and unadjusted estimates with different covariate sets. When ORs and 95% CIs were not reported, they were calculated directly from the reported genotype or carrier counts for that study. A continuity correction (0.5 added to each cell) was applied if any cell in the  $2 \times 2$  table contained a zero to allow computation of the OR and its variance.

## 2.6. Synthesis methods

Study-specific ORs were log-transformed, and their standard errors were used to pool effects in a random-effects meta-analysis, with primary analyses conducted using a random-effects model with variance estimated by restricted maximum likelihood (REML) in Stata (StataCorp, College Station, TX), consistent with recommendations for heterogeneous observational data [24,25]. Between-study heterogeneity was evaluated using Cochran's Q test and quantified with the  $I^2$  statistic, and Galbraith plots were used to identify potential outliers and explore sources of heterogeneity. As a sensitivity analysis, we also fitted fixed-effect models using the Mantel-Haenszel method and compared these

pooled estimates with the REML-based results. Subgroup analyses were conducted according to ancestry and other available secondary outcomes where data permitted. Additional sensitivity analyses were performed to examine the stability of the pooled effect under different sample-size thresholds ( $n > 300$ ,  $n > 500$ ,  $n > 1000$ ), by splitting studies into large ( $\geq 785$  participants) and small ( $< 785$  participants) samples using the median total sample size as the cut-point, and by excluding studies with extreme CI bounds (upper 95% CI  $> 500$ ). Publication bias and small-study effects were assessed using funnel plots and Egger's regression test. All tests were two-sided, with a significance level of 0.05.

## 2.7. Rationale for post hoc changes

All core methods followed the prespecified PROSPERO protocol (CRD420251182022). After inspecting the included studies, we introduced additional, exploratory sensitivity analyses that were not explicitly detailed in the protocol. Many studies were small and had large standard errors, leading to high within-study variance. Under these conditions, the Cochran Q statistic tends to approximate its degrees of freedom, yielding  $I^2$  values close to zero even when true between-study differences may exist, because between-study variance is difficult to detect against the large within-study variance. To assess the robustness of the pooled effect under these circumstances, we added post hoc

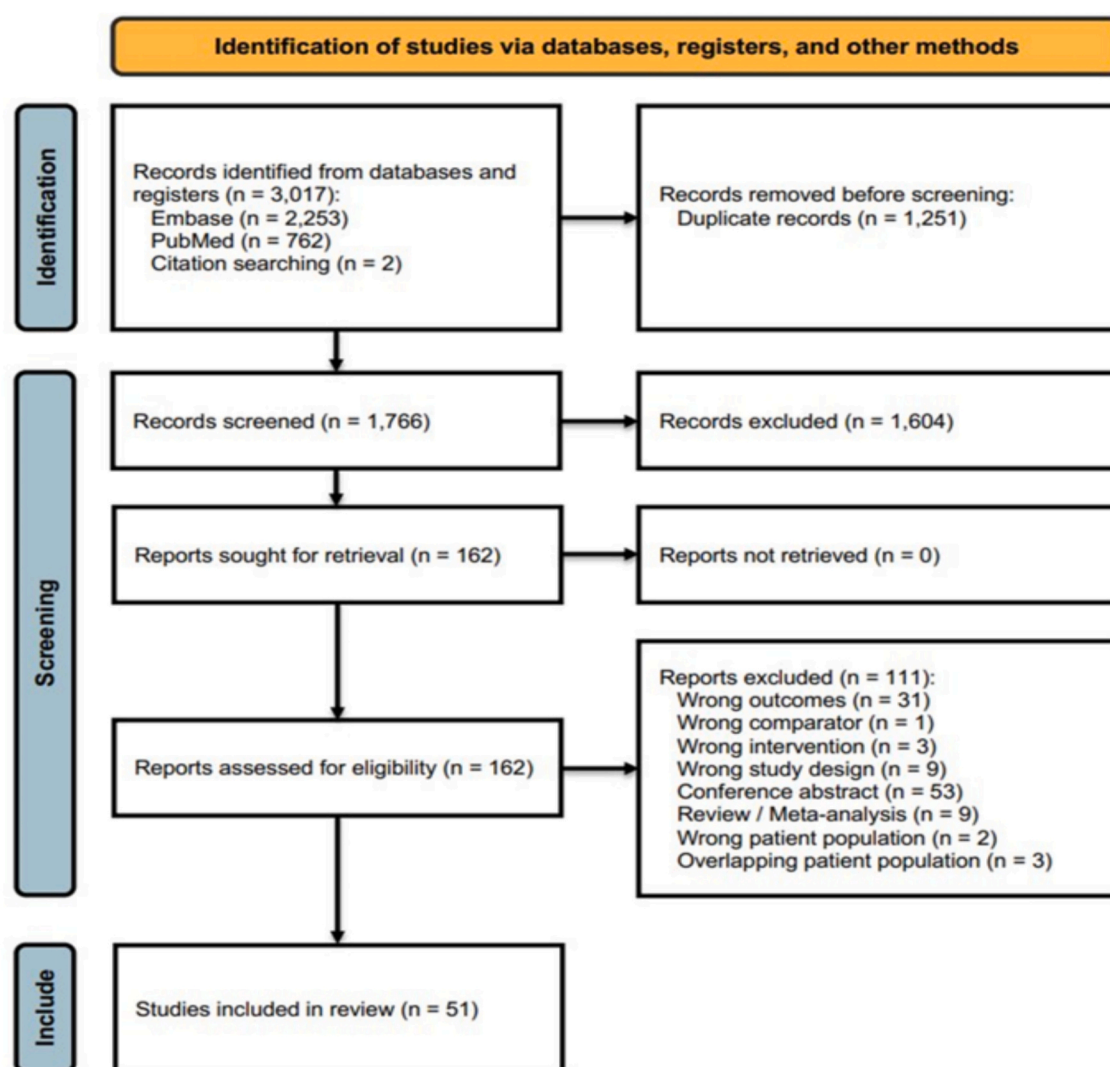


Fig. 1. PRISMA flow diagram of study selection process.

sensitivity analyses using sample size thresholds ( $n > 300$ ,  $n > 500$ ,  $n > 1000$ ), splitting studies at the median total sample size, excluding studies with extreme CI bounds (upper 95% CI  $> 500$ ), and comparing random-effects with fixed-effect Mantel–Haenszel models. These analyses were undertaken solely to evaluate the influence of highly imprecise studies and did not alter the primary conclusions.

### 3. Results

#### 3.1. Study selection

We identified 3017 records from Embase ( $n = 2253$ ), PubMed ( $n = 762$ ), and citation searching ( $n = 2$ ). After removing 1251 duplicates, 1766 unique records were screened by title and abstract. Of these, 162 full-text articles were assessed for eligibility, and 51 studies met the inclusion criteria and were included in the systematic review and meta-analysis. The full study selection process is shown in the PRISMA flow diagram (Fig. 1).

#### 3.2. Evidence table

The characteristics of all included studies are summarized in Table 1. The descriptors in this table illustrate the diversity of contributing cohorts across countries and ancestral groups, as well as the variation in study size and quality.

#### 3.3. Study characteristics

In total, 51 studies, including 33,752 cases and 34,101 controls focused on the effect of L444P and PD [11,15,16,26–73].

#### 3.4. Risk of bias

Across all 51 studies, NOS scores ranged from 5 to 9, with the majority ( $n = 45$ ) scoring between 7 and 9, suggesting low risk of bias. Total score distribution is shown in Supplementary Fig. 1, where most studies cluster at the upper end of the scale, and only a small subset fall within the moderate-risk range.

Domain-level patterns are summarized by Supplementary Fig. 2. Most studies achieved high scores in the selection and outcome/exposure domains, reflecting robust case definitions, appropriate control selection, and consistent genotyping procedures. The comparability domain exhibited the most variation, as many studies did not adjust for potential confounders (e.g., age, sex, ancestry), consistent with the reliance on unadjusted carrier frequency data in genetic association research.

#### 3.5. Main analyses

In the primary random-effects meta-analysis (REML) of 51 studies, the pooled OR for PD among heterozygous *GBA1* L444P carriers versus non-carriers was 9.19 (95% CI 6.94–12.16). Formal heterogeneity statistics suggested no between-study heterogeneity ( $\tau^2 = 0.00$ ;  $I^2 = 0\%$ ;  $Q(50) = 18.60$ ,  $p = 1.00$ ), a pattern expected when many studies have large standard errors and between-study variance is hard to detect statistically despite plausible clinical or methodological differences. This has motivated the additional sensitivity analyses described below. Study-specific and pooled estimates are shown in the forest plot (Fig. 2).

#### 3.6. Sensitivity analyses

To assess the robustness of the association between heterozygous *GBA1* L444P carrier status and PD, we conducted a series of sensitivity analyses varying study inclusion criteria and meta-analytic methods (summarized in Supplementary Table 1).

##### 3.6.1. Leave-one-out meta-analysis

We examined the influence of each study on the pooled effect by iteratively removing one study at a time and refitting the random-effects model. Across all 51 iterations, the pooled ORs remained stable, ranging from 8.70 to 9.47 compared with the primary estimate of 9.19 (95% CI 6.94–12.16). No individual study meaningfully altered the magnitude or direction of the association (Supplementary Fig. 3), indicating that the overall result was not driven by any single influential study.

##### 3.6.2. Cumulative meta-analysis

A cumulative meta-analysis, ordered by year of publication, showed that early studies tended to have imprecise and highly variable estimates, but the pooled OR increased rapidly as additional studies were incorporated and stabilized beginning around 2008–2010. From that point onward, the cumulative pooled OR remained consistently in the range of approximately 7 to 9, with narrow CIs as evidence accumulated (Supplementary Fig. 4). This pattern indicates that the association between heterozygous *GBA1* L444P carrier status and PD risk has been stable over time as the body of evidence has expanded.

##### 3.6.3. Sensitivity analyses based on study size

We evaluated whether the association differed when restricting the meta-analysis to larger studies. Limiting the analysis to studies with more than 300 participants (44 studies) resulted in a pooled OR of 9.56 (95% CI 7.16–12.77). When the threshold was increased to more than 500 participants (34 studies), the pooled OR was 10.05 (95% CI 7.39–13.66). Restricting to the largest studies with more than 1000 participants (18 studies) yielded a pooled OR of 10.70 (95% CI 7.48–15.29). Across all three thresholds, the effect estimates were similar in magnitude to the primary analysis and remained well above the null, indicating that the strong association was not driven solely by smaller studies. The corresponding forest plots are shown in Supplementary Figs. 5.1–5.3.

##### 3.6.4. Split by study size

To evaluate whether the association differed according to overall study size, we stratified studies into “large” (total sample size  $\geq 785$  participants; 26 studies) and “small” ( $< 785$  participants; 25 studies). In the large-study subgroup, the pooled OR was 10.14 (95% CI 7.33–14.03). In the small-study subgroup, the pooled OR was 6.90 (95% CI 3.97–11.99). Although the point estimate was somewhat higher in larger studies, the CIs overlapped substantially and a formal test of subgroup differences was not statistically significant ( $Q_b(1) = 1.39$ ,  $p = 0.24$ ). These findings indicate that the elevated risk of PD among L444P carriers is consistent across studies of different sizes and is not driven by small-study effects. The corresponding forest plot is shown in Supplementary Fig. 6.

##### 3.6.5. Extreme confidence interval exclusion

To assess the impact of very imprecise studies, we excluded four studies with extremely wide CIs (defined as an upper 95% CI  $> 500$ ). The resulting meta-analysis of 48 studies gave a pooled OR of 9.02 (95% CI 6.79–11.99), almost identical to the primary estimate. This indicates that the overall association is not driven by studies with extreme uncertainty. The corresponding forest plot is shown in Supplementary Fig. 7.

##### 3.6.6. Fixed-effect (Mantel–Haenszel) sensitivity analyses

As an additional model-based sensitivity analysis, we applied a fixed-effect Mantel–Haenszel approach to estimate the pooled association between heterozygous *GBA1* L444P carrier status and PD. The Mantel–Haenszel pooled OR was 10.35 (95% CI 7.80–13.73), which was very similar to the REML random-effects estimate. This consistency indicates that the overall findings are robust to the choice of meta-analytic model and are not driven by assumptions about between-study variance. The full Mantel–Haenszel forest plot is shown in Supplementary Fig. 8.



**Table 1**Characteristics of included studies examining the association between *GBA1* L444P and PD.

| Study               | Year | Country        | Ancestry         | Group* | Sample size (case/control) | Genotype method                | NOS |
|---------------------|------|----------------|------------------|--------|----------------------------|--------------------------------|-----|
| Asselta             | 2014 | Italy          | West Eurasian    | Non AJ | 2766/1111                  | Targeted mutation assays       | 8   |
| Benitez             | 2016 | USA            | West Eurasian    | Non AJ | 478/337                    | Short-read NGS                 | 8   |
| Biswas              | 2021 | India          | South Asian      | Non AJ | 334/241                    | Targeted mutation assays       | 7   |
| Bras                | 2009 | Portugal       | West Eurasian    | Non AJ | 230/430                    | Sanger sequencing (full exons) | 8   |
| Choi                | 2012 | South Korea    | East Asian       | Non AJ | 277/291                    | Sanger sequencing (full exons) | 8   |
| Clark               | 2007 | USA            | West Eurasian    | Non AJ | 102/94                     | Sanger sequencing (full exons) | 8   |
| Crosiers            | 2016 | Belgium        | West Eurasian    | Non AJ | 266/536                    | Sanger sequencing (full exons) | 7   |
| De Marco            | 2008 | Italy          | West Eurasian    | Non AJ | 395/483                    | Targeted mutation assays       | 7   |
| De Melo Amaral      | 2019 | Brazil         | Admixed American | Non AJ | 81/81                      | Targeted mutation assays       | 8   |
| den Heijer          | 2020 | Netherlands    | West Eurasian    | Non AJ | 3402/655                   | Pseudogene-aware / long-range  | 6   |
| Duran               | 2013 | United Kingdom | West Eurasian    | Non AJ | 185/283                    | Sanger sequencing (full exons) | 8   |
| Emelyanov           | 2018 | Russia         | West Eurasian    | Non AJ | 762/400                    | Targeted mutation assays       | 7   |
| Gan-Or              | 2008 | Israel         | West Eurasian    | AJ     | 420/4138                   | Targeted mutation assays       | 8   |
| González-del Rincón | 2013 | Mexico         | Admixed American | Non AJ | 128/252                    | Targeted mutation assays       | 8   |
| Guimarães           | 2011 | Brazil         | Admixed American | Non AJ | 347/341                    | Targeted mutation assays       | 8   |
| Guo                 | 2015 | China          | East Asian       | Non AJ | 1019/1030                  | Targeted mutation assays       | 9   |
| Han                 | 2015 | Canada         | West Eurasian    | Non AJ | 225/110                    | Sanger sequencing (full exons) | 7   |
| Hassanin            | 2025 | UK/Luxembourg  | West Eurasian    | Non AJ | 653/767                    | Pseudogene-aware / long-range  | 9   |
| Huang               | 2011 | Taiwan         | East Asian       | Non AJ | 967/780                    | Targeted mutation assays       | 8   |
| Kalinderi           | 2009 | Greece         | West Eurasian    | Non AJ | 172/132                    | Sanger sequencing (full exons) | 5   |
| Kumar               | 2012 | Serbia         | West Eurasian    | Non AJ | 360/348                    | Sanger sequencing (full exons) | 8   |
| Lesage              | 2011 | France         | West Eurasian    | Non AJ | 1130/391                   | Sanger sequencing (full exons) | 8   |
| Li                  | 2014 | Japan          | East Asian       | Non AJ | 147/100                    | Sanger sequencing (full exons) | 6   |
| Mao                 | 2010 | China          | East Asian       | Non AJ | 616/411                    | Targeted mutation assays       | 8   |
| Mata                | 2008 | USA            | West Eurasian    | Non AJ | 721/554                    | Targeted mutation assays       | 8   |
| Mitsui              | 2009 | Japan          | East Asian       | Non AJ | 534/544                    | Sanger sequencing (full exons) | 8   |
| Moraitou            | 2011 | Greece         | West Eurasian    | Non AJ | 205/206                    | Targeted mutation assays       | 7   |
| Muldmaa             | 2021 | Estonia        | West Eurasian    | Non AJ | 189/158                    | Short-read NGS                 | 8   |
| Neumann             | 2009 | United Kingdom | West Eurasian    | Non AJ | 790/257                    | Targeted mutation assays       | 7   |
| Noreau              | 2011 | Canada         | West Eurasian    | Non AJ | 212/189                    | Sanger sequencing (full exons) | 8   |
| Pachchek            | 2023 | Luxembourg     | West Eurasian    | Non AJ | 637/675                    | Pseudogene-aware / long-range  | 8   |
| Pakarulrazy         | 2020 | Malaysia       | East Asian       | Non AJ | 100/50                     | Short-read NGS                 | 7   |
| Pulkes              | 2014 | Thailand       | East Asian       | Non AJ | 482/396                    | Sanger sequencing (full exons) | 7   |
| Ran                 | 2016 | Sweden         | West Eurasian    | Non AJ | 1625/2025                  | Short-read NGS                 | 7   |
| Ruskey              | 2019 | USA_Israel     | West Eurasian    | AJ     | 735/662                    | Short-read NGS                 | 9   |
| Sato                | 2005 | Canada         | West Eurasian    | Non AJ | 88/122                     | Targeted mutation assays       | 8   |
| Setó-Salvia         | 2011 | Spain          | West Eurasian    | Non AJ | 225/186                    | Sanger sequencing (full exons) | 8   |
| Spitz               | 2008 | Brazil         | Admixed American | Non AJ | 65/267                     | Targeted mutation assays       | 6   |
| Straniero           | 2020 | Italy          | West Eurasian    | Non AJ | 3832/7757                  | Pseudogene-aware / long-range  | 6   |
| Sun                 | 2010 | China          | East Asian       | Non AJ | 402/413                    | Targeted mutation assays       | 8   |
| Tan                 | 2007 | Singapore      | East Asian       | Non AJ | 331/347                    | Targeted mutation assays       | 8   |
| Toft                | 2006 | Norway         | West Eurasian    | Non AJ | 311/474                    | Targeted mutation assays       | 8   |
| Török               | 2016 | Hungary        | West Eurasian    | Non AJ | 124/122                    | Targeted mutation assays       | 8   |
| Vélez-Pardo         | 2019 | Colombia/Peru  | Admixed American | Non AJ | 602/319                    | Pseudogene-aware / long-range  | 8   |
| Wang                | 2012 | China          | East Asian       | Non AJ | 208/298                    | Targeted mutation assays       | 8   |
| Wu                  | 2007 | Taiwan         | East Asian       | Non AJ | 518/339                    | Targeted mutation assays       | 8   |
| Ylönen              | 2017 | Finland        | West Eurasian    | Non AJ | 849/403                    | Targeted mutation assays       | 6   |
| Yu                  | 2015 | China          | East Asian       | Non AJ | 184/130                    | Sanger sequencing (full exons) | 8   |
| Zhang               | 2012 | China          | East Asian       | Non AJ | 195/443                    | Targeted mutation assays       | 8   |
| Zhou                | 2023 | China          | East Asian       | Non AJ | 4034/2931                  | Short-read NGS                 | 8   |
| Ziegler             | 2007 | Taiwan         | East Asian       | Non AJ | 92/92                      | Sanger sequencing (full exons) | 8   |

\* AJ = Ashkenazi Jews.

### 3.7. Subgroup analyses & meta regression results

#### 3.7.1. Subgroup analyses by ancestry

To evaluate whether the association between heterozygous *GBA1* L444P carrier status and PD varied across populations, we conducted ancestry-specific subgroup analyses. Ancestry categories were defined using broad continental groupings commonly applied in human genetic epidemiology – West Eurasian, East Asian, South Asian, and admixed American – following the population structure framework described by Risch et al., in which human genetic variation clusters into a small number of major continental groups [74]. Since ancestry was assigned based on study-reported categories and broad continental groupings, some misclassification within these groups is possible.

Across ancestry groups, the association remained consistently elevated. Among West Eurasian studies ( $n = 29$ ), the pooled OR was 7.92 (95% CI 5.60–11.19). Among East Asian studies ( $n = 16$ ), the pooled OR was 11.83 (95% CI 7.05–19.86). For admixed American populations ( $n = 5$ ), the pooled estimate was 15.78 (95% CI 4.22–58.96), although CIs

were wider due to smaller sample sizes. Only one South Asian study was available, yielding an OR of 8.06 (95% CI 0.39–168.19).

CIs across ancestry groups overlapped, and the formal test of subgroup differences was not statistically significant ( $Q_{b(3)} = 2.28$ ,  $p = 0.52$ ). The ancestry-specific forest plot is shown in Supplementary Fig. 9.

#### 3.8. Evidence of publication bias

Visual inspection of the funnel plot showed an approximately symmetric distribution of study estimates around the pooled effect, with no marked absence of small studies on either side of the mean. The Galbraith plot showed most studies falling within the 95% confidence region, without a pattern suggestive of systematic small-study effects or outliers driving the association. Consistent with the graphical assessments, the Egger regression test did not provide evidence of small-study effects ( $\beta = -0.30$ ,  $SE = 0.33$ ,  $z = -0.93$ ,  $p = 0.35$ ). Taken together, these results indicate no statistically significant evidence of publication bias in this meta-analysis. The funnel plot and Galbraith plot are shown

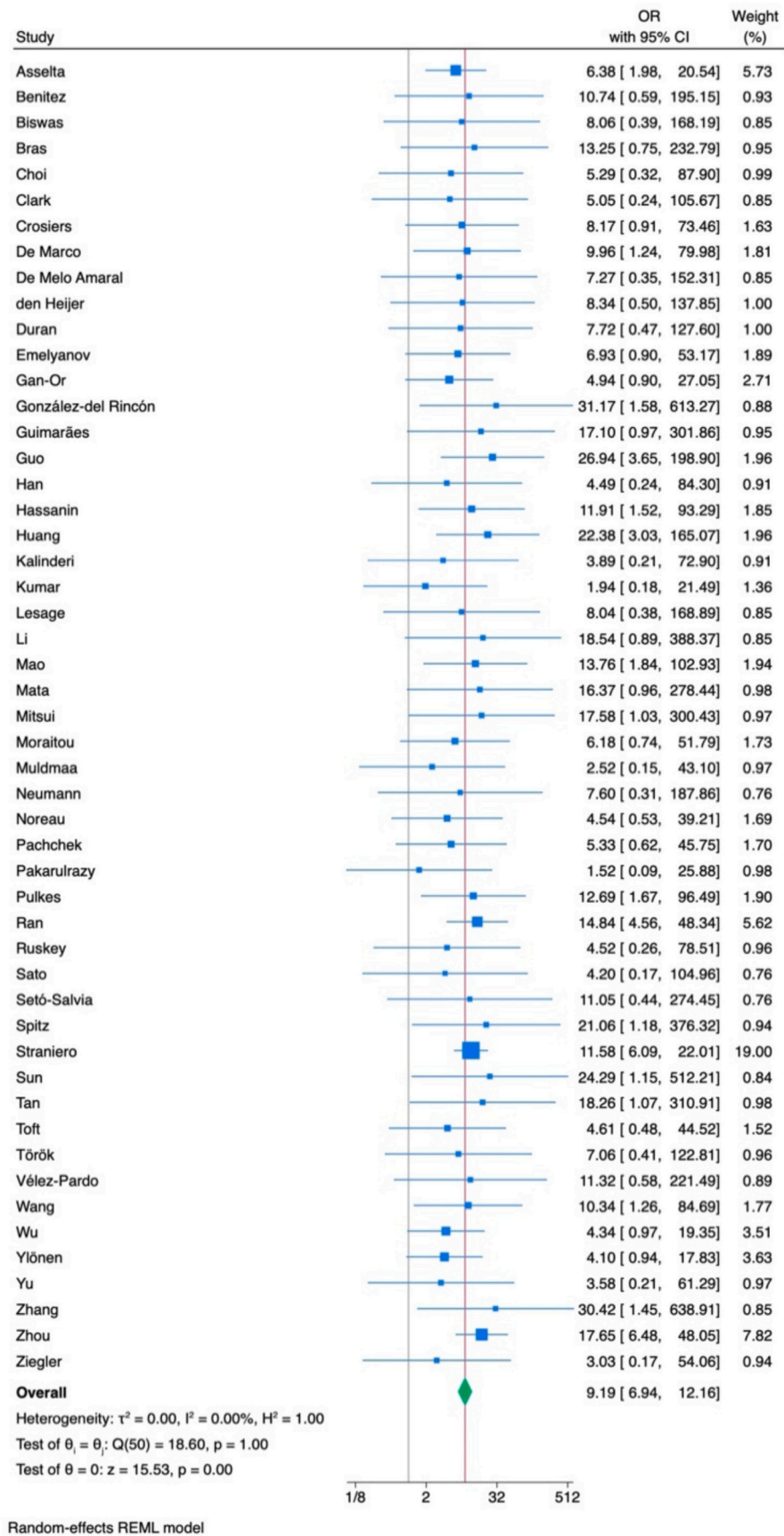


Fig. 2. Forest plot of the association between heterozygous *GBA1* L444P status and PD.

in Supplementary Figs. 10 and 11.

#### 4. Discussion

This review represents the largest synthesis to date focused specifically on heterozygous *GBA1* L444P and PD, pooling 51 studies to provide the most precise and current risk estimate for this severe variant. Our pooled OR (~9–10) is broadly consistent with previous meta-analyses [20,21]. Elevated ORs were observed across West Eurasian, East Asian, and admixed American populations, with overlapping confidence intervals and no significant ancestry-specific differences. The magnitude and consistency of this association are biologically plausible, as L444P is a severe *GBA1* mutation associated with markedly reduced glucocerebrosidase activity, lysosomal dysfunction implicated in  $\alpha$ -synuclein accumulation, and faster disease progression [75].

Importantly, our findings apply specifically to heterozygous L444P carriers and should not be extrapolated to the full spectrum of *GBA1* variants, which exhibit substantial heterogeneity in penetrance and phenotypic expression. Broader mutation-inclusive analyses remain essential to fully characterize *GBA1*-associated PD.

Recent biochemical evidence supports the biological relevance of severe *GBA1* variants in heterozygous carriers. Szymańska-Rózek et al. showed that carriers of the L444P variant exhibit persistently elevated levels of glucosylsphingosine (lyso-Gb1), a marker of lysosomal dysfunction, compared with carriers of milder *GBA1* variants, with elevations persisting into adulthood [76]. These findings reinforce the importance of variant severity in *GBA1*-associated PD and provide a mechanistic link between sustained lysosomal dysfunction and the magnitude of risk observed in our meta-analysis.

Building on earlier syntheses, we systematically explored heterogeneity using ancestry-stratified analyses, multiple sample-size thresholds, leave-one-out and cumulative meta-analyses, and both random- and fixed-effect (Mantel–Haenszel) models. Although conventional heterogeneity statistics ( $I^2 = 0\%$ ) likely underestimate true between-study variability because many included studies were small and imprecise, all of these sensitivity analyses yielded similar pooled effect estimates, supporting the robustness of the association despite limited power to detect heterogeneity.

Several limitations should be considered when interpreting these findings. First, although formal heterogeneity statistics ( $I^2 = 0\%$ ) suggested no between-study heterogeneity, this likely reflects limited power given the large number of small, imprecise studies and wide confidence intervals in several subgroups. These patterns reflect sparse data and underrepresentation in PD genetics and limit our ability to fully characterize true between-study and within-ancestry variability, even though extensive sensitivity analyses consistently produced similar effect estimates [9]. Consistent with Higgins & Thompson, non-significant Q tests and low  $I^2$  values are common when studies are small or imprecise, so the absence of statistical heterogeneity does not imply true homogeneity [77]. Second, our meta-analysis relied primarily on unadjusted ORs derived from *GBA1* L444P carrier counts to maximize comparability across studies, many of which did not report multivariable models. Therefore, residual confounding by age, sex, ancestry, or recruitment settings might influence the magnitude of the observed association. However, as *GBA1* L444P is a germline variant and thus not modified by age, sex, or clinical status, these factors are more likely to modestly inflate or attenuate the estimated OR than to generate a spurious large effect or reverse the direction of association. The consistently elevated odds of PD in L444P carriers across diverse studies and ancestry groups and the overlapping confidence intervals suggest that residual confounding alone is unlikely to explain the observed increase in risk. Third, although we used a large language model-based assistant as a supplementary tool to flag potential errors or inconsistencies, we implemented safeguards to minimize automation bias: LLM outputs were never accepted without verification, and all final data and quality ratings were based on human review. LLM-assisted

workflows should therefore be viewed as an adjunct, not a replacement, for conventional multi-reviewer methods. Third, we focused exclusively on heterozygous L444P carriers and had sparse or no data for some ancestry groups (e.g., South Asian, African populations), leading to imprecise subgroup estimates and limiting the generalizability of our findings across populations and *GBA1* variants. Broad continental ancestry categories are inherently coarse and may not capture finer-scale genetic diversity. Accordingly, apparent differences in point estimates across ancestry subgroups should be interpreted cautiously, as overlapping confidence intervals and limited sample sizes preclude firm conclusions regarding true ancestry-specific variation in effect size.

Finally, while the NOS provided a structured assessment of study quality, it has recognized limitations, including subjective domain scoring and unequal weighting. NOS scores were applied descriptively rather than for analytic weighting. Although most included studies were rated moderate to high quality, residual bias not captured by the NOS cannot be fully excluded.

An additional limitation relates to the complex genomic structure of the *GBA1* locus. *GBA1* has a highly homologous pseudogene (*GBAP1*), and the L444P (p.Leu483Pro) variant lies within a region prone to gene conversion and recombinant allele formation [78]. Recombinant alleles incorporating pseudogene sequence may include L444P together with additional variants in cis. Genotyping methods varied across included studies and included targeted panels and sequencing-based approaches, which differ in their ability to resolve recombinant or complex alleles. As our analysis relied on published genotype calls, we were unable to systematically distinguish isolated L444P variants from recombinant or cis-complex alleles. This potential misclassification represents a limitation and should be considered when interpreting the findings.

PD associated with *GBA1* variants was among the first observations to challenge the long-standing assumption that heterozygous carriers of lysosomal storage disorder genes are clinically unaffected. While *GBA*–PD initially appeared to represent a gene-specific exception, emerging genetic and epidemiological evidence suggests a broader paradigm in which heterozygous variation across multiple lysosomal genes are associated with adult-onset neurodegenerative risk. Recent work, including that by Benítez and colleagues, supports the view that lysosomal dysfunction may represent a shared biological substrate linking rare metabolic disorders with more common late-onset neurological phenotypes [79].

Within this evolving framework, *GBA1*-associated PD has been conceptualized as a genetically defined clinico-biological entity characterized by earlier onset, more aggressive progression, and increased non-motor burden compared to idiopathic PD. The term “Sidransky syndrome” has recently been proposed to reflect this coherence and to emphasize the continuum between Gaucher disease and *GBA1*-related PD [80,81]. This framing underscores the importance of recognizing *GBA1*-related PD as part of the broader lysosomal–neurodegenerative continuum rather than as an isolated genetic association.

Overall, these results support routine consideration of *GBA1* L444P in PD risk counseling, call for broader and more equitable access to high-quality genetic testing, and prioritize improving estimate precision in underrepresented populations through large, population-based studies and focused investigation of severe *GBA1* variants. Overall, this literature supports a gradual shift in how lysosomal dysfunction is understood in PD, while acknowledging substantial heterogeneity, and provides a foundation for future precision-medicine approaches that integrate variant severity, polygenic background, and lysosomal pathway involvement [82,83].

#### CRedit authorship contribution statement

**Agnieszka Jurecka:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Hyun Kyung Kim:** Writing – review & editing, Writing – original draft,

Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Noelle Siegfried:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yu Chen Zhao:** Writing – review & editing, Supervision, Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ymgme.2026.110112>.

## Data availability

No data was used for the research described in the article.

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